

{ M.f → interaction
text
image
clustering
PCA }

Recsys using a supervised learning

Practical session

NetFlix

Movies

Users

Recsys :-

- i) Supervised Learning
- ii) Matrix factorization
- iii) Evaluation

n_1	n_2	(target (ratings))	Rating?
User / features	Movie / features		

1 year $\left(\begin{matrix} 9 \text{ months} \rightarrow \text{Train} \\ 3 \text{ months} \rightarrow \text{Test} \end{matrix} \right) \rightarrow \text{learn}$

Should we do Random split or sequential split for Recommendation Systems?

★ (Sequential split) ★ given your dataset has any time based column.

(Recsys / Time series forecasting)

How to decide in ML?

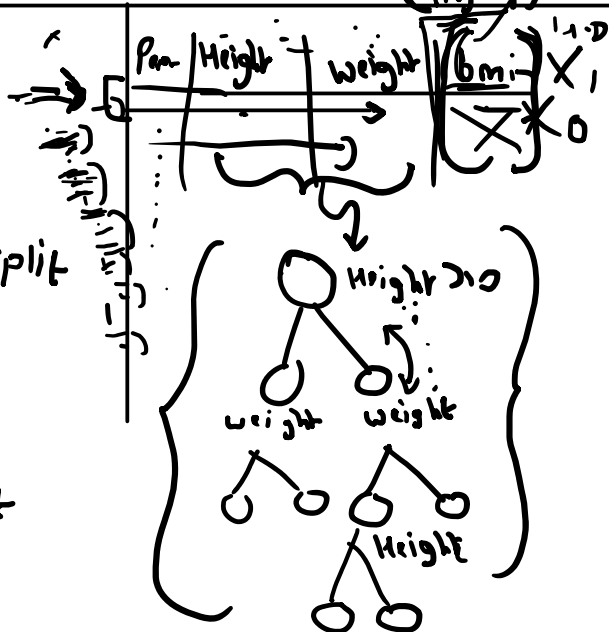
evaluate Model $\rightarrow$ future. Training data $\rightarrow$ past data. Testing data $\rightarrow$ future.

When aggregated historical info is used to create features, always do sequential split.

★ (Aggregate feature) ★ $\left(\begin{matrix} \text{count} \\ \text{sum} \\ \text{mean} \\ \text{median} \end{matrix} \right)$ (groupby) $\rightarrow$ NO. of movies rated by users. (Non linear) XGBoost. Linear Regression.

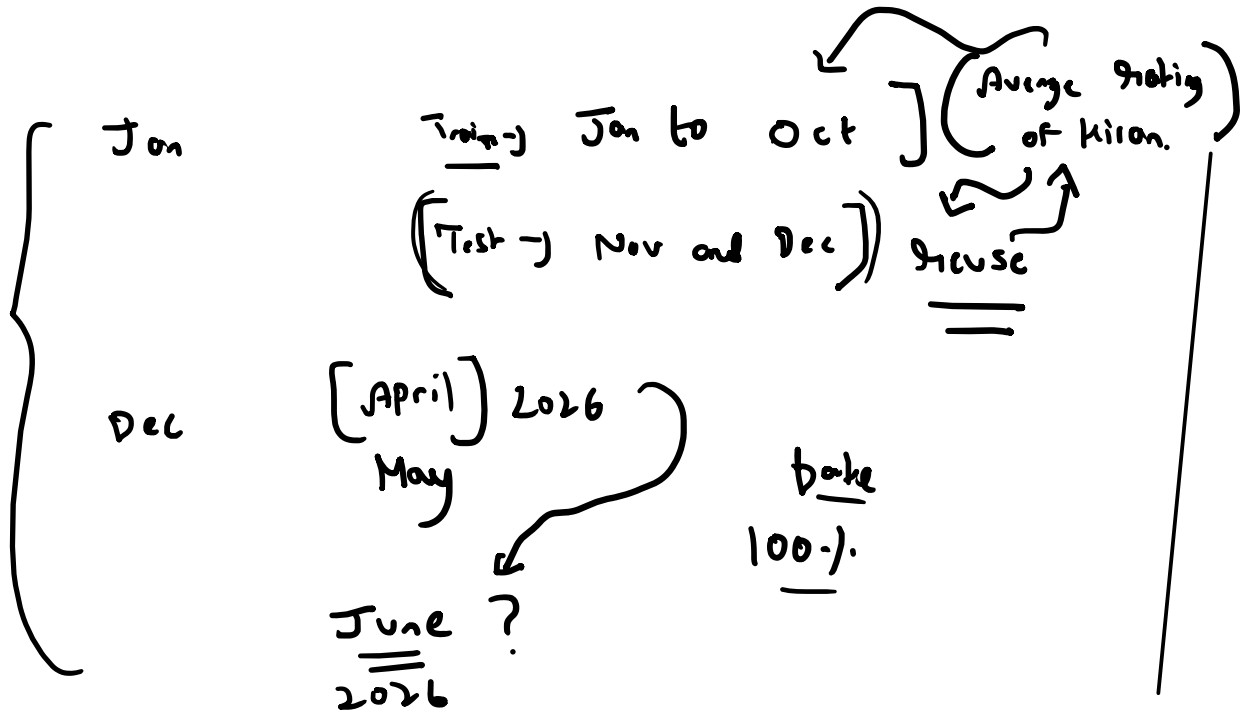
Case 1: Create agg features then train-test sequential split

Case 2: First train test split. Create agg features on training set. Join features to test set



★ ① Train-test split, then create features ★ $\left(\begin{matrix} \text{Past (Train)} \\ \text{Future (Test)} \end{matrix} \right)$

2) Check feats, train-test split



0.9

RMSE $\rightarrow$ Measures Mean error in "Rating" value

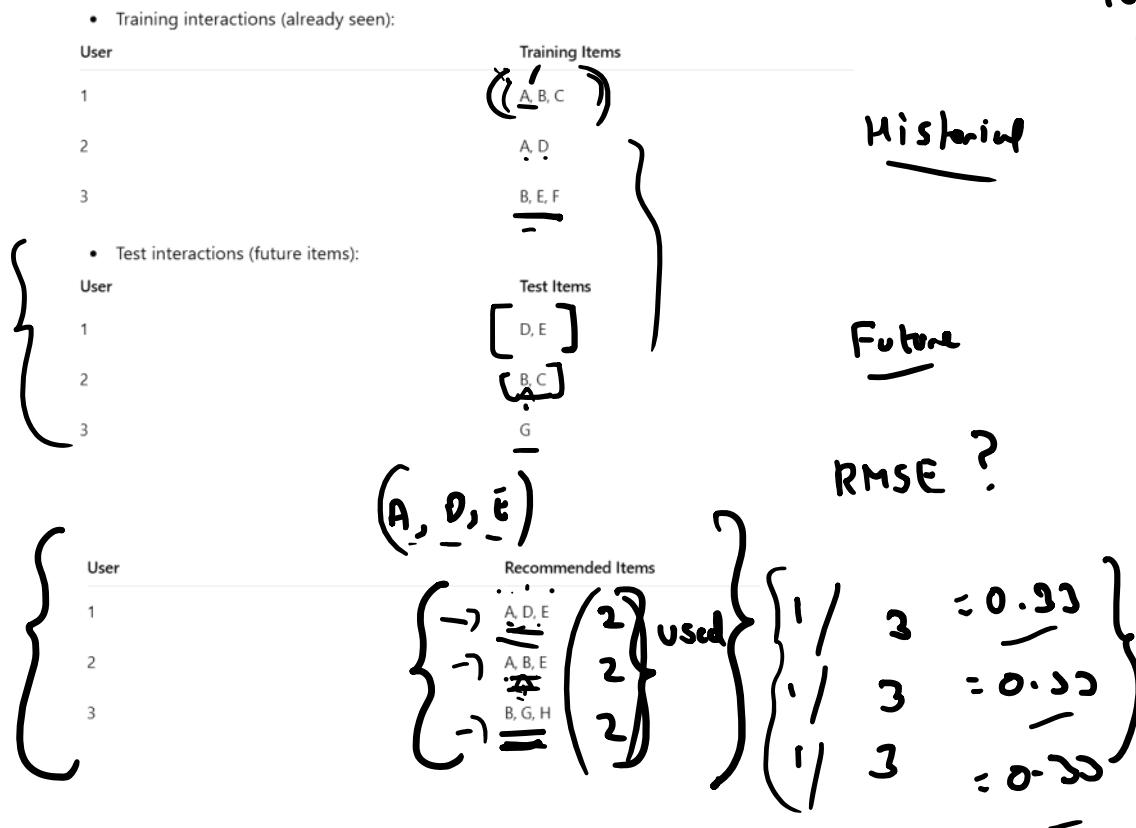
It does not tell us Quality of Recommendations

(M.F or S or Hybrid)

Uses Item

Sort ratings

Using which items have top ratings
Top 2 to recommend



$\frac{33}{100} \rightarrow$ Average = 0.33 $\rightarrow$ k-overlap
67 new prob.

User	Recommended	Training	Overlap@3	Seen Hits
1	A, D, E	A, B, C	1 / 3 = 0.33	1 (A)
2	A, B, E	A, D	1 / 3 = 0.33	1 (A)
3	B, G, H	B, E, F	1 / 3 = 0.33	1 (B)

Average Overlap@3 (training) = $(0.33 + 0.33 + 0.33)/3 \approx 0.33$

Out of all predictions
how many were a success

User	Recommended	Test Items	Precision@3	Hits
1	A, D, E	D, E	$2/3 \approx 0.67$	2 (D, E)
2	A, B, E	B, C	$1/3 \approx 0.33$	1 (B)
3	B, G, H	G	$1/3 \approx 0.33$	1 (G)

in the future

Average Precision@3 = $(0.67 + 0.33 + 0.33)/3 \approx 0.44$

Precision @ k

ULF
U1 2

ILF
I1 1
I2 2
I3 2.5
I4 0.5
I5 2

Predict.
 $ULF \times ILF = \text{Rating}$

Top 3 items

U1	I1	2
U1	I2	4
U1	I3	5
U1	I4	1
U1	I5	4

Recommendations

I1,
I2,
I5